

PRAVAAH

Disaster Management

Intelligent System

Flood Prediction, Prioritization, Emergency Planning & Resource Allocation

PRESENTED BY:

Team D-Coders

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PROBLEM STATEMENT

Problem Title – Intelligent Disaster Response for Flood Prediction, Emergency Planning & Resource Allocation

What does this Problem Aim to Solve?

When a flood is predicted, response teams face a critical question: *which locality should get help first with limited resources?* Decisions are often made on the spot, slowing response and risking misallocation. The problem is to bridge the gap between *flood prediction and actionable response.*

Why did your team choose this?

Flooding is one of India's most frequent and damaging disasters. Existing systems often stop at predicting flood risk. We wanted to turn that prediction into a *ranked, actionable, resource-aware emergency plan.*

REAL-WORLD PROBLEM ALIGNMENT

Who faces this issue?



Officials



Rescue Teams

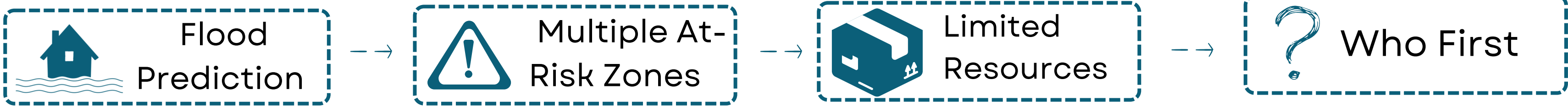


Volunteers



Citizens

What's the current gap in the system?



How does your proposed solution fill that gap practically?



PROPOSED SOLUTION

What are you Building?

We are building *PRAVAAH* - a Disaster Response Intelligence Platform that predicts flood risk, prioritizes localities, and recommends actionable response plans for better resource allocation.

It converts raw flood predictions into explainable priority scores and data-driven emergency plans for officials, rescue teams, and communities.

What are the 3-4 Key Features?



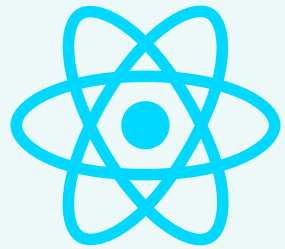
What makes it different or innovative?



TECH STACK & ARCHITECTURE

Tech Stack

FRONTEND



React, Vite,
Tailwind

BACKEND



Go (Golang)
API

ML/AI



Python
Sci-kit learn

MAPS



Leaflet.js
Open Street
Map

APIs/Libraries

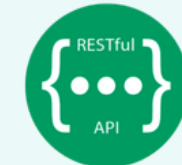
- Scikit - Learn
- Pandas
- NumPy
- Leaflet
- OpenStreetMap
- Go net/http



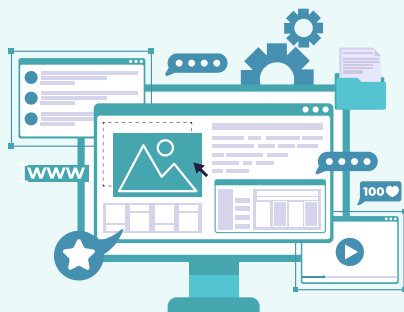
Pandas



Leaflet



Deployment Plan



Develop and Test



Build and containerize
(DOCKER)



Deploy on Cloud

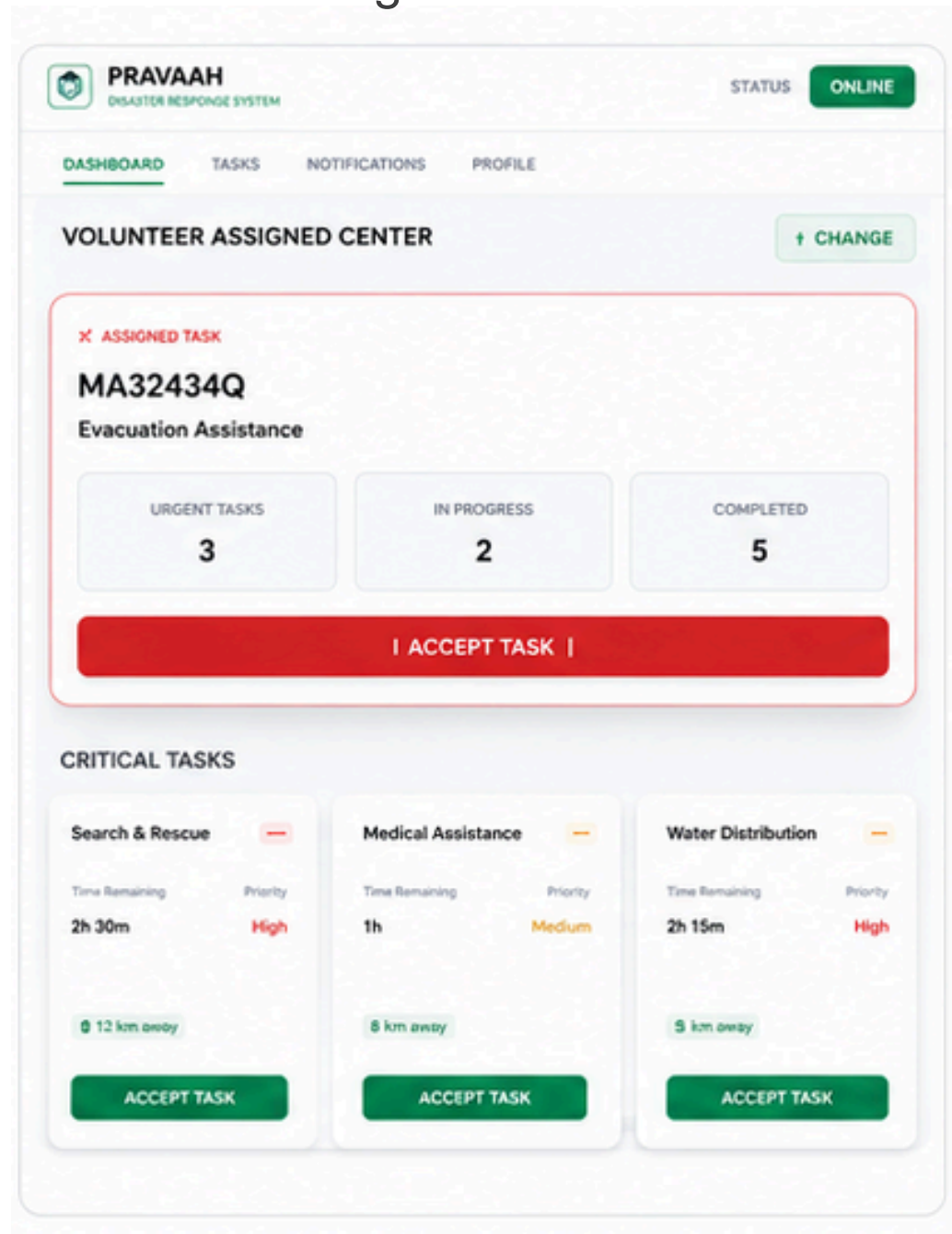


Live Application
(Accessible to User)

PROTOTYPE DEMO

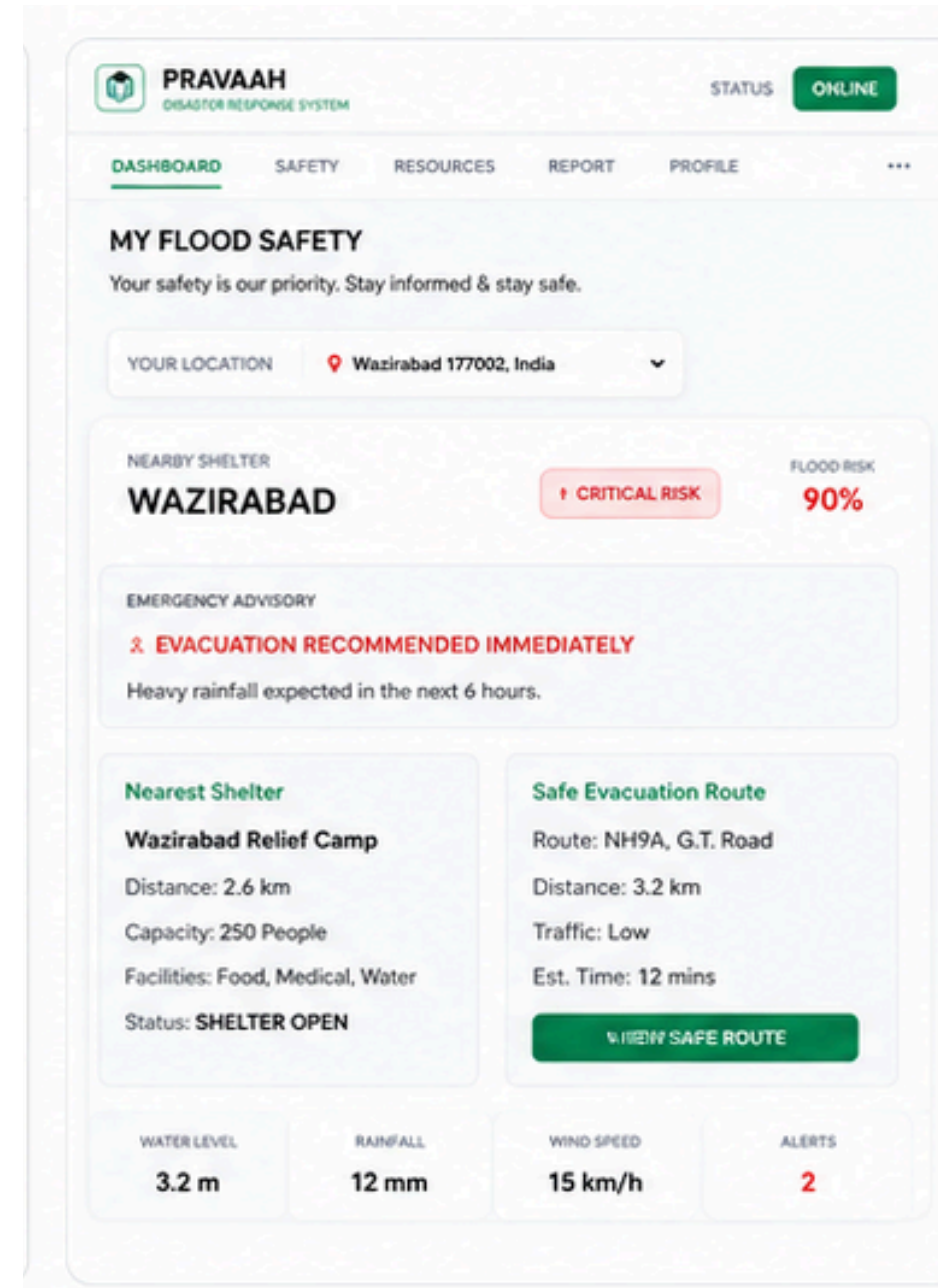
1. Volunteer Task Dashboard

Volunteers can view & accept tasks based on urgent needs.



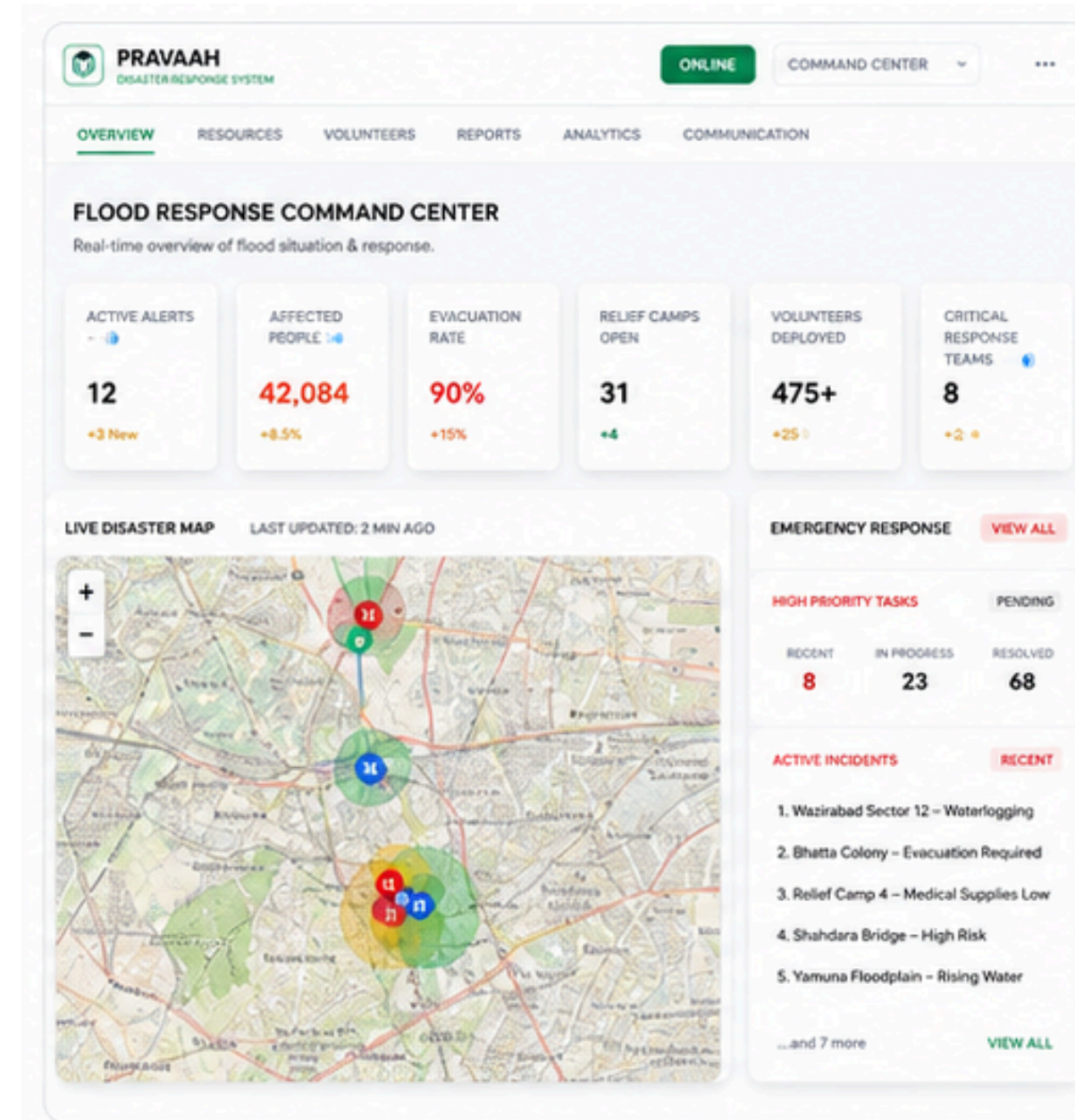
2. Citizen Safety Portal

Citizens get risk alerts, safe routes, shelter locations & live advisories.



3. Official Command Center

Officials get overview, live map, AI recommendations & resource allocation support.



4. Links

Here are the links to the resources

[Drive Link](#)

[Github](#)